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Subject- Agile Software and DevOps Lab

Experiment 02

AIM: To containerise the application using docker

LO4: Containerize, orchestrate, and deploy applications using Docker, Docker Compose, Kubernetes, and cloud infrastructure tools.

INSTALLATION STEPS:

1. **Enable Virtualization**
   Enable **Intel VT-x** or **AMD-V** from the BIOS/UEFI settings to support Docker containers.
2. **Install WSL 2**
   Open PowerShell as Administrator and run:

wsl --install

Restart your computer after installation.

3. **Download and Install Docker Desktop**
   Download Docker Desktop from the official Docker website and complete the installation using the default settings (WSL 2 recommended).
4. **Launch Docker Desktop**
   Open Docker Desktop and wait until it starts successfully.
5. **Verify the Installation**
   Run the following commands in Command Prompt or PowerShell:
6. docker --version

docker run hello-world

If the **"Hello from Docker!"** message is displayed, Docker has been installed successfully and is ready to use.

THEORY:

1. What is Docker?

Docker is an **open-source containerization platform** that enables developers to build, package, distribute, and run applications in isolated environments called **containers**. It bundles an application with all its dependencies, libraries, runtime, and configuration files into a single package, ensuring that the application runs consistently across different systems such as a developer's computer, testing environment, or production server.

Unlike traditional deployment methods, Docker eliminates the common issue of "It works on my machine" by creating identical execution environments. Docker is lightweight because it shares the host operating system's kernel instead of running a complete operating system for every application.

Features of Docker

- Lightweight and fast
- Platform-independent
- Consistent development and deployment
- Easy application portability
- Efficient resource utilization
- Simplified software distribution
- Supports microservices architecture

Advantages

- Faster application deployment
- Reduced infrastructure costs
- Easy scalability
- Better resource management
- Version-controlled images
- Simplified collaboration among developers

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2. What is Containerization?

Containerization is a virtualization technique in which an application and all of its dependencies are packaged into a single unit called a **container**. This container can be executed on any system that has Docker installed, regardless of the underlying operating system or hardware.

Containers isolate applications from one another while sharing the host operating system kernel. This makes them significantly lighter and faster than traditional virtual machines.

In containerization, each container contains:

- Application source code
- Runtime environment
- Required libraries
- Dependencies
- Configuration files
- Environment variables

Working of Containerization

1. Developer writes the application.
2. A Dockerfile is created to define the environment.
3. Docker builds an image.
4. The image is stored locally or pushed to Docker Hub.
5. Containers are created from the image.
6. The application runs identically on any machine.

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3. What is a Docker Image?

A **Docker Image** is a **read-only template** that contains everything required to run an application. It includes the operating system files, application code, runtime, libraries, dependencies, configuration files, and startup instructions.

Images are created using a **Dockerfile**, which contains a series of instructions that Docker follows to build the image.

An image itself does not execute; instead, it acts as a blueprint from which one or more containers can be created.

Characteristics of Docker Images

- Read-only
- Immutable (cannot be modified after creation)
- Can be versioned using tags
- Portable across systems
- Reusable

- Stored locally or on Docker Hub

Image Layers

Docker images are built using multiple layers:

- Base Operating System
- Language Runtime
- Dependencies
- Application Code
- Configuration
- Startup Command

Each layer is cached, making image builds faster and more efficient.

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4. What is a Docker Container?

A **Docker Container** is a running instance of a Docker image. It is an isolated environment in which an application executes with all of its required resources.

Containers are lightweight because they share the host operating system kernel instead of running their own operating system.

Each container has:

- Its own file system
- Network interface
- Process space
- Environment variables
- Running application

---

5. Difference Between Docker Image and Docker Container

Docker Image	**Docker Container**
Blueprint of an application	Running instance of an image
Read-only	Read-write
Static	Dynamic
Cannot execute by itself	Executes the application
Created using Dockerfile	Created from an image

Docker Image	Docker Container
Can create multiple containers	Exists only while running or until removed
Stored in Docker Hub or local system	Stored in Docker Engine while running

COMMANDS

1. Version : Checks the version of docker installed

```
C:\Users\15L>docker --version
Docker version 25.0.3, build 4debf41

C:\Users\15L>docker images
REPOSITORY    TAG       IMAGE ID       CREATED          SIZE
nginx         latest    fd204fe2f750   4 months ago     161MB
hello-world   latest    1b44b5a3e06a   11 months ago    10.1kB
ubuntu        latest    ca2b0f26964c   2 years ago      77.9MB
```

2. Docker search ubuntu - - filter stars= 500 : Gives the images that are equal to that number or more

```
C:\Users\15L>docker search ubuntu
NAME                              DESCRIPTION                                     STARS     OFFICIAL
ubuntu                            Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid                      Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx                      Nginx, a high-performance reverse proxy & we…   141
ubuntu/bind9                      BIND 9 is a very flexible, full-featured DNS…   120
ubuntu/kafka                      Apache Kafka, a distributed event streaming …   62
ubuntu/zookeeper                  ZooKeeper maintains configuration informatio…   15
ubuntu/apache2                    Apache, a secure & extensible open-source HT…   104
ubuntu/prometheus                 Prometheus is a systems and service monitori…   80
ubuntu/jre                        Chiseled Java runtime based on Ubuntu. Long-…   25
ubuntu/dotnet-aspnet              Chiselled Ubuntu runtime image for ASP.NET a…   26
ubuntu/python                     A chiselled Ubuntu rock with the Python runt…   33
ubuntu/mysql                      MySQL open source fast, stable, multi-thread…   74
ubuntu/postgres                   PostgreSQL is an open source object-relation…   45
ubuntu/dotnet-runtime             Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/redis                      Redis, an open source key-value store. Long-…   26
ubuntu/dotnet-deps                Chiselled Ubuntu for self-contained .NET & A…   16
ubuntu/grafana                    Grafana, a feature rich metrics dashboard & …   12
ubuntu/memcached                  Memcached, in-memory keyvalue store for smal…   7
ubuntu/jdk                        Ubuntu rock for OpenJDK                         0
ubuntu/rabbitmq                   Ubuntu-based RabbitMQ image - an open source…   0
ubuntu/mimir                      Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/loki                       Grafana Loki, a log aggregation system like …   2
ubuntu/traefik                    Ubuntu ROCK for Traefik, a modern HTTP rever…   3
ubuntu/opentelemetry-collector    This is a rock for the OpenTelemetry Collect…   0
ubuntu/alertmanager               Ubuntu Rock for Alertmanager. It handles ale…   1
```

```
C:\Users\15L>docker search ubuntu --filter "stars=500"
NAME      DESCRIPTION                                     STARS     OFFICIAL
ubuntu    Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
```

3. Docker pull ubuntu: Downloads images from docker

```
ubuntu/python                   A chiselled Ubuntu rock with the Python runt…   33
ubuntu/mysql                    MySQL open source fast, stable, multi-thread…   74
ubuntu/postgres                 PostgreSQL is an open source object-relation…   45
ubuntu/redis                    Redis, an open source key-value store. Long-…   26
ubuntu/dotnet-runtime           Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/dotnet-deps              Chiselled Ubuntu for self-contained .NET & A…   16
ubuntu/grafana                  Grafana, a feature rich metrics dashboard & …   12
ubuntu/memcached                Memcached, in-memory keyvalue store for smal…   7
ubuntu/jdk                      Ubuntu rock for OpenJDK                         0
ubuntu/rabbitmq                 Ubuntu-based RabbitMQ image - an open source…   0
ubuntu/mimir                    Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/loki                     Grafana Loki, a log aggregation system like …   2
ubuntu/traefik                  Ubuntu ROCK for Traefik, a modern HTTP rever…   3
ubuntu/opentelemetry-collector  This is a rock for the OpenTelemetry Collect…   0
ubuntu/alertmanager             Ubuntu Rock for Alertmanager. It handles ale…   1

C:\Users\15L>docker search ubuntu --filter stars=500
NAME      DESCRIPTION                                     STARS     OFFICIAL
ubuntu    Ubuntu is a Debian-based Linux operating sys…   17859     [OK]

C:\Users\15L>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
ed819469700f: Pull complete
a3679419df18: Pull complete
Digest: sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
```

4. Docker run ubuntu /bin/bash: creates a new Ubuntu container and opens an interactive Bash terminal inside it
5. Docker ps- Displays a list of all currently running Docker containers.

```
C:\Users\15L>docker images
REPOSITORY    TAG       IMAGE ID       CREATED         SIZE
ubuntu        latest    de7345b16e94   6 days ago      100MB
nginx         latest    fd204fe2f750   4 months ago    161MB
hello-world   latest    1b44b5a3e06a   11 months ago   10.1kB
ubuntu        <none>    ca2b0f26964c   2 years ago     77.9MB

C:\Users\15L>docker run ubuntu/bin/bash
Unable to find image 'ubuntu/bin/bash:latest' locally
docker: Error response from daemon: pull access denied for ubuntu/bin/bash, repository does not exist or may require 'do
cker login': denied: requested access to the resource is denied.
See 'docker run --help'.

C:\Users\15L>docker run ubuntu /bin/bash

C:\Users\15L>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

C:\Users\15L>docker run -it
"docker run" requires at least 1 argument.
See 'docker run --help'.

Usage:  docker run [OPTIONS] IMAGE [COMMAND] [ARG...]

Create and run a new container from an image

C:\Users\15L>docker run -it ubuntu /bin/bash
root@d5e54f781411:/#
```

6. Docker run: Create and start a new container from a Docker image.

```
C:\Users\15L>docker run --name mycontainer -it ubuntu bash
root@f1d175be2472:/# exit
exit
```

7. Docker stop: Sends a termination signal to the container, allowing the application inside the container to shut down properly before the container stops.

```
C:\Users\15L>docker stop mycontainer
mycontainer
```

8. Docker start:Used to restart a stopped container.

```
C:\Users\15L>docker start mycontainer
mycontainer
```

9. Docker inspect :Used to display detailed information about a Docker object, such as a container, image, network, or volume.



```
            "WorkingDir": "",
            "Entrypoint": null,
            "OnBuild": null,
            "Labels": {
                "org.opencontainers.image.created": "2026-07-13T16:06:30.499069+00:00",
                "org.opencontainers.image.description": "The Ubuntu container image maintained by Canonical\n\nUbuntu is a Debian-based Linux operating system that runs from the desktop to the cloud, to all yo
ur internet connected things.\nIt is the world's most popular operating system across public clouds and OpenStack clouds.\nIt is the number one platform for containers; from Docker to Kubernetes to LXD, Ubuntu
 can run your containers at scale.\nFast, secure and simple, Ubuntu powers millions of PCs worldwide.\n",
                "org.opencontainers.image.title": "ubuntu",
                "org.opencontainers.image.version": "26.04"
            }
        },
        "NetworkSettings": {
            "Bridge": "",
            "SandboxID": "8b0dd6144bb90f7e585f2b66fd3f11adbcf94bc74e50d85585f283c8bd4d6a8e",
            "SandboxKey": "/var/run/docker/netns/8b0dd6144bb9",
            "Ports": {},
            "HairpinMode": false,
            "LinkLocalIPv6Address": "",
            "LinkLocalIPv6PrefixLen": 0,
            "SecondaryIPAddresses": null,
            "SecondaryIPv6Addresses": null,
            "EndpointID": "730350d5da4b961daefb500c7803e4a37565a1a56cb740a570fea5e34f822aa6",
            "Gateway": "172.17.0.1",
            "GlobalIPv6Address": "",
            "GlobalIPv6PrefixLen": 0,
            "IPAddress": "172.17.0.2",
            "IPPrefixLen": 16,
            "IPv6Gateway": "",
            "MacAddress": "02:42:ac:11:00:02",
            "Networks": {
                "bridge": {
                    "IPAMConfig": null,
                    "Links": null,
                    "Aliases": null,
                    "MacAddress": "02:42:ac:11:00:02",
                    "NetworkID": "3c22b4b58101914cad4e7ab53cf301e4e4499bfc14e7bf47f2bf8ab058333b14",
                    "EndpointID": "730350d5da4b961daefb500c7803e4a37565a1a56cb740a570fea5e34f822aa6",
                    "Gateway": "172.17.0.1",
                    "IPAddress": "172.17.0.2",
                    "IPPrefixLen": 16,
                    "IPv6Gateway": "",
                    "GlobalIPv6Address": "",
                    "GlobalIPv6PrefixLen": 0,
                    "DriverOpts": null,
                    "DNSNames": null
                }
            }
        }
    }
]
```


10. Docker ps -a: Shows all containers

```
C:\Users\15L>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED         STATUS              PORTS     NAMES
f1d175be2472   ubuntu    "bash"    4 minutes ago   Up About a minute             mycontainer

C:\Users\15L>docker ps -a
CONTAINER ID   IMAGE                              COMMAND                  CREATED         STATUS                     PORTS                    NAMES
f1d175be2472   ubuntu                             "bash"                   4 minutes ago   Up 2 minutes                                        mycontainer
3f9eed2969de   ubuntu                             "/bin/bash"              12 minutes ago  Exited (127) 4 minutes ago                          modest_perlman
16eccf1a92e9   ubuntu                             "/bin/bash"              15 minutes ago  Exited (0) 15 minutes ago                           kind_diffie
e8841e749448   sample-flask-app:v1                "flask run --host=0.…"   4 months ago    Created                                             stupefied_brattain
a169e1386175   sample-flask-app:v1                "flask run --host=0.…"   4 months ago    Exited (255) 3 months ago  0.0.0.0:5000->5000/tcp   funny_newton
8e37dff4d72c   sample-flask-app:v1                "."                      4 months ago    Created                    0.0.0.0:5000->5000/tcp   angry_feistel
41ceff137f19   nginx                              "/docker-entrypoint.…"   4 months ago    Exited (255) 3 months ago  0.0.0.0:3000->80/tcp     quizzical_saha
05652b53729f   httpd                              "httpd-foreground"       4 months ago    Exited (255) 3 months ago  0.0.0.0:9090->80/tcp     focused_cerf
171bbef15299   nginx                              "/docker-entrypoint.…"   4 months ago    Created                                             objective_nobel
68e8ba592c7f   nginx                              "/docker-entrypoint.…"   4 months ago    Created                                             trusting_feistel
48f8e7a8ca1b   nginx                              "/docker-entrypoint.…"   4 months ago    Created                                             fervent_jepsen
3872caa94e58   nginx                              "/docker-entrypoint.…"   4 months ago    Created                                             practical_banzai
341cff10e221   ca2b0f26964c                       "bash"                   4 months ago    Exited (127) 4 months ago                           hungry_napier
3da66845e08b   hello-world                        "/hello"                 4 months ago    Exited (0) 4 months ago                             magical_heyrovsky
4f85ce4bf575   httpd                              "httpd-foreground"       2 years ago     Exited (255) 4 months ago  80/tcp                   determined_bohr
b34f2765cc66   ca2b0f26964c                       "/bin/bash"              2 years ago     Exited (0) 2 years ago                              reverent_euclid
72f2f13072f3   welcome-to-docker:latest           "docker-entrypoint.s…"   2 years ago     Exited (255) 2 years ago   0.0.0.0:8089->3000/tcp   competent_hawking
b30cf9e8fa7e   welcome-to-docker:latest           "docker-entrypoint.s…"   2 years ago     Created                                             zen_chaplygin
8dd6df41521d   docker/welcome-to-docker:latest    "/docker-entrypoint.…"   2 years ago     Exited (255) 2 years ago   0.0.0.0:8088->80/tcp     welcome-to-docker
```

11. Docker stats: Display real-time resource usage statistics of running containers.

```
C:\Users\15L>docker stats
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
^C
```

12. Docker rm: Removes container

```
C:\Users\15L>docker rm -f mycontainer
mycontainer
```

13.Run nginx in the background

```
C:\Users\15L>docker run -d --name web nginx
bc50f0a51bb67bf8b8b9db6f2a7f2667598c075ffd6ad886e4d4e9af7ce6a865
```

Linux Commands: Used to perform operations and manage files, directories, and processes inside a Linux system.

```
root@d5e54f781411:/# pwd
/
root@d5e54f781411:/# ls
bin   dev  home  lib64  mnt  proc  run   srv  tmp  var
boot  etc  lib   media  opt  root  sbin  sys  usr
root@d5e54f781411:/# mkdir test
root@d5e54f781411:/# cd test
root@d5e54f781411:/test# touch file1.txt
root@d5e54f781411:/test# ls
file1.txt
```

14. Docker Pull Alpine: Used to download the Alpine Linux image from Docker Hub.

```
C:\Users\15L>docker pull alpine
Using default tag: latest
latest: Pulling from library/alpine
55afa1ecc21d: Pull complete
Digest: sha256:28bd5fe8b56d1bd048e5babf5b10710ebe0bae67db86916198a6eec43
4943f8b
Status: Downloaded newer image for alpine:latest
docker.io/library/alpine:latest

What's Next?
  1. Sign in to your Docker account → docker login
  2. View a summary of image vulnerabilities and recommendations → docke
r scout quickview alpine

C:\Users\15L>
```

15. Search for Image: Used to search and find available Docker images from Docker Hub using the docker search command.

```
C:\Users\15L>docker search python
NAME                      DESCRIPTION                                     STARS     OFFICIAL
python                    Python is an interpreted, interactive, objec…   10488     [OK]
circleci/python           Python is an interpreted, interactive, objec…   116
cimg/python                                                               25
demisto/python            The Demisto base python image                   5
ubuntu/python             A chiselled Ubuntu rock with the Python runt…   33
okteto/python                                                             0
paketobuildpacks/python                                                   0
cleanstart/python         Deploy, scale, and secure critical infrastru…   1
activestate/python        ActiveState's customizable, low-to-no vulner…   11
intel/python                                                              5
corpusops/python          https://github.com/corpusops/docker-images/     1
rootpublic/python                                                         0
gentkit/python            A Docker images of Python environment .         0
stagex/python             High-level scripting language                   0
linuxserver/python                                                        0
vulhub/python                                                             2
koxudaxi/python           Dockerfiles for CPython of lysnikolaou's tag…   0
unidata/python            Container with Unidata's python packages as …   3
clarinpl/python                                                           0
dtcenter/python           Necessary Python3 packages for plotting GRIB…   1
openeuler/python                                                          1
dragonflyoss/python                                                       0
nanozoo/python                                                            0
hkube/python                                                              0
python/docker-bpo         Docker image for local development of http:/…   0

C:\Users\15L>
```

16. Docker Image Prune: Used to remove unused Docker images and free up storage space.

```
C:\Users\15L>docker image prune
WARNING! This will remove all dangling images.
Are you sure you want to continue? [y/N] y
Total reclaimed space: 0B
```

17. Docker Container LS: Used to list all currently running Docker containers.

```
C:\Users\15L>docker container ls -a
CONTAINER ID   IMAGE                              COMMAND                  CREATED          STATUS                      PORTS                    NAMES
3f9eed2969de   ubuntu                             "/bin/bash"              26 minutes ago   Exited (127) 19 minutes ago                          modest_perlman
16eccf1a92e9   ubuntu                             "/bin/bash"              30 minutes ago   Exited (0) 30 minutes ago                            kind_diffie
e8841e749448   sample-flask-app:v1                "flask run --host=0.…"   4 months ago     Created                                              stupefied_brattain
a169e1386175   sample-flask-app:v1                "flask run --host=0.…"   4 months ago     Exited (255) 3 months ago   0.0.0.0:5000->5000/tcp   funny_newton
8e37dff40d72c  sample-flask-app:v1                "."                      4 months ago     Created                     0.0.0.0:5000->5000/tcp   angry_feistel
41ceff137f19   nginx                              "/docker-entrypoint.…"   4 months ago     Exited (255) 3 months ago   0.0.0.0:3000->80/tcp     quizzical_saha
05652b53729f   httpd                              "httpd-foreground"       4 months ago     Exited (255) 3 months ago   0.0.0.0:9090->80/tcp     focused_cerf
171bbef15299   nginx                              "/docker-entrypoint.…"   4 months ago     Created                                              objective_nobel
68e8ba592c7f   nginx                              "/docker-entrypoint.…"   4 months ago     Created                                              trusting_feistel
48f8e7a8ca1b   nginx                              "/docker-entrypoint.…"   4 months ago     Created                                              fervent_jepsen
3872caa94e58   nginx                              "/docker-entrypoint.…"   4 months ago     Created                                              practical_banzai
341cff10e221   ca2b0f26964c                       "bash"                   4 months ago     Exited (127) 4 months ago                            hungry_napier
3da66845e08b   hello-world                        "/hello"                 4 months ago     Exited (0) 4 months ago                              magical_heyrovsky
4f85ce4bf575   httpd                              "httpd-foreground"       2 years ago      Exited (255) 4 months ago   80/tcp                   determined_bohr
b34f2765cc66   ca2b0f26964c                       "/bin/bash"              2 years ago      Exited (0) 2 years ago                               reverent_euclid
72f2f13072f3   welcome-to-docker:latest           "docker-entrypoint.s…"   2 years ago      Exited (255) 2 years ago    0.0.0.0:8089->3000/tcp   competent_hawking
b30cf9e8fa7e   welcome-to-docker:latest           "docker-entrypoint.s…"   2 years ago      Created                                              zen_chaplygin
8dd6df41521d   docker/welcome-to-docker:latest    "/docker-entrypoint.…"   2 years ago      Exited (255) 2 years ago    0.0.0.0:8088->80/tcp     welcome-to-docker
```

18. Docker Info: Used to display detailed information about the Docker system and configuration.

```
C:\Users\15L>docker info
Client:
 Version:    25.0.3
 Context:    default
 Debug Mode: false
 Plugins:
  buildx: Docker Buildx (Docker Inc.)
    Version:  v0.12.1-desktop.4
    Path:     C:\Program Files\Docker\cli-plugins\docker-buildx.exe
  compose: Docker Compose (Docker Inc.)
    Version:  v2.24.6-desktop.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-compose.exe
  debug: Get a shell into any image or container. (Docker Inc.)
    Version:  0.0.24
    Path:     C:\Program Files\Docker\cli-plugins\docker-debug.exe
  dev: Docker Dev Environments (Docker Inc.)
    Version:  v0.1.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-dev.exe
  extension: Manages Docker extensions (Docker Inc.)
    Version:  v0.2.22
    Path:     C:\Program Files\Docker\cli-plugins\docker-extension.exe
  feedback: Provide feedback, right in your terminal! (Docker Inc.)
    Version:  v1.0.4
    Path:     C:\Program Files\Docker\cli-plugins\docker-feedback.exe
  init: Creates Docker-related starter files for your project (Docker Inc.)
    Version:  v1.0.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-init.exe
  sbom: View the packaged-based Software Bill Of Materials (SBOM) for an image (Anchore Inc.)
    Version:  0.6.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-sbom.exe
  scout: Docker Scout (Docker Inc.)
    Version:  v1.5.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-scout.exe

Server:
 Containers: 6
  Running: 0
  Paused: 0
  Stopped: 6
 Images: 5
 Server Version: 25.0.3
 Storage Driver: overlay2
  Backing Filesystem: extfs
  Supports d_type: true
  Using metacopy: false
  Native Overlay Diff: true
  userxattr: false
 Logging Driver: json-file
 Cgroup Driver: cgroupfs
```

19. Docker History: Used to show the history and layers of a Docker image.

```
C:\Users\15L>docker history ubuntu
IMAGE          CREATED      CREATED BY                                      SIZE      COMMENT
de7345b16e94   6 days ago   umoci raw add-layer --image /home/buildd/roc…   179B      Add rock control metadata
<missing>      6 days ago   umoci config --image /home/buildd/rockcraft-…   0B        Set annotations
<missing>      6 days ago   umoci config --image /home/buildd/rockcraft-…   0B        Set labels
<missing>      6 days ago   umoci config --image /home/buildd/rockcraft-…   0B        Set default PATH for bare-based rock
<missing>      6 days ago   umoci config --image /home/buildd/rockcraft-…   0B        Set default commands
<missing>      6 days ago   umoci config --image /home/buildd/rockcraft-…   0B        Set entrypoint
<missing>      6 days ago   umoci raw add-layer --image /home/buildd/roc…   100MB
```

20. Disk Usage (docker system df): Used to display the amount of disk space used by Docker images, containers, and volumes.

```
C:\Users\15L>docker system df
TYPE            TOTAL     ACTIVE    SIZE      RECLAIMABLE
Images          5         4         347.3MB   8.416MB (2%)
Containers      6         0         1.127kB   1.127kB (100%)
Local Volumes   0         0         0B        0B
Build Cache     1         0         0B        0B
```

21. Docker Network Inspect: Used to view detailed information about a Docker network.

```
C:\Users\15L>docker network inspect bridge
[
    {
        "Name": "bridge",
        "Id": "3c22b4b58101914cad4e7ab53cf301e4e4499bfc14e7bf47f2bf8ab058333b14",
        "Created": "2026-07-20T09:49:36.630387228Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": [
                {
                    "Subnet": "172.17.0.0/16",
                    "Gateway": "172.17.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Containers": {},
        "Options": {
            "com.docker.network.bridge.default_bridge": "true",
            "com.docker.network.bridge.enable_icc": "true",
            "com.docker.network.bridge.enable_ip_masquerade": "true",
            "com.docker.network.bridge.host_binding_ipv4": "0.0.0.0",
            "com.docker.network.bridge.name": "docker0",
            "com.docker.network.driver.mtu": "1500"
        },
        "Labels": {}
    }
]
```

22. Docker Volume: Used to create and manage persistent storage for Docker containers.

```
C:\Users\15L>docker volume create myvolume
myvolume
```

23. Docker Network: Used to create and manage communication between Docker containers.

```
C:\Users\15L>docker network create mynetwork
1c4b0027ac879f19bfb71b0c302affd27ef570546ef2ddbec28aadce3ef36481
```

24. Docker System Prune: Used to remove unused containers, images, networks, and cache to free Docker storage space.

```
C:\Users\15L>docker system prune
WARNING! This will remove:
  - all stopped containers
  - all networks not used by at least one container
  - all dangling images
  - unused build cache

Are you sure you want to continue? [y/N] y
Deleted Containers:
3f9eed2969de6505e99ec9a3d8a4d6b2fb17589f8fc2eb3de09847b45a516802
16eccf1a92e902c9006ffdbac81cccdb32967eb934a97034bbc57880fb27c650
e8841e7494487745be27a6deeab286cf0b378aebe11cf792fa57c88a3884affe1
a169e1386175350ada9d3e82fd34f926b8cf9c135b6e418becbf9d165b09ecc5
8e37dff4d72cc38c35e8195ed7c0d2fb79fcdef2c274ca6cfff456995ab82bca
41ceff137f195e952ea5e7578a6eefbf19a9f943ccda5c9fd3a9c85ec4b01db3
05652b53729f8d7612955ec3d27a696d60b140ddafd31336ecf0b91e81b613f8
171bbef152997528fe8db286bf431d2a7dbe441457e69c987011aff0815f19ec
68e8ba592c7f25bcddcf8080f2b9c2401e9735f1635374cb888897573eebefdd
48f8e7a8ca1be7e2c5a6aac5c4d6d7300c21dbd4017ed9b407d4aa6b303ca34e
3872caa94e589e2e6eb2323fc28f6689bb8f0cdfe0d4e6e701cadc0176d3fef0
341cff10e22189b92964fa064cabca3784c8110725714621cfb38e31def84c71
3da66845e08b4eb2c5303c47d07c260f9b5e43c8e78d1e951876dfa36034d76b
4f85ce4bf575093d972e1617eab716a3d6997c3b6ccd95b721ae3539ab213c99
b34f2765cc66fc34f86dc58d113ac1811ce5518b9812500a10339dcd979c2e5d
72f2f13072f302b0edb24abe7681c52ba7be1fab1566b7c668f5c52940cb75e9
b30cf9e8fa7e17a8db8974e5d1194c5e98c081e0b245baf9291f6919311f9f29
8dd6df41521d3bcab04928f2bc8e0c1e38f97e2a05e09fd6bfabc82c9c0a7a2a

Deleted Networks:
mynetwork

Deleted Images:
untagged: ubuntu@sha256:77906da86b60585ce12215807090eb327e7386c8fafb5402369e421f44eff17e
deleted: sha256:ca2b0f26964cf2e80ba3e084d5983dab293fdb87485dc6445f3f7bbfc89d7459
deleted: sha256:5498e8c22f6996f25ef193ee58617d5b37e2a96decf22e72de13c3b34e147591

Deleted build cache objects:
su2b2pnjuh36bspucwm6zmjx1
thvp6halgjg06vzoqpg2f4e5r
k2p81999forouopsh0ck7p9fi
9iqhg73dfjup9soqq2ivunuo8
oqs50qlok4a6ta4v3ii7j2wz0
31ewzdd14f00g9rvf5fivsrbc
mhc2jq30psx1xfqq85ewy2yx2

Total reclaimed space: 78.73MB
```

25. Docker rename

```
C:\Users\15L>docker rename b125f8a6a583 ubuntu-test

C:\Users\15L>docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS                      PORTS
    NAMES
b125f8a6a583   ubuntu         "/bin/bash"              33 minutes ago   Exited (0) 33 minutes ago
    ubuntu-test
5fde9bfdd162   ca2b0f26964c   "bash -c 'echo Hello…"   4 months ago     Exited (0) 4 months ago
    frosty_kepler
6587c9c4a2bf   nginx          "/docker-entrypoint.…"   4 months ago     Exited (0) 4 months ago
    charming_franklin
adefa716cb32   hello-world    "/hello"                 4 months ago     Exited (0) 4 months ago
    jolly_swanson
77ea50afe062   ca2b0f26964c   "bash -c 'echo Hello…"   4 months ago     Exited (0) 4 months ago
    agitated_agnesi
56c98c2698d7   ca2b0f26964c   "bash"                   4 months ago     Exited (0) 4 months ago
    pedantic_golick
```

CONCLUSION:

In this experiment, we learned how Docker simplifies application deployment by packaging the application along with its dependencies into containers. We understood the concept of containerization and the difference between Docker images and containers. Overall, Docker provides a lightweight, consistent, and efficient environment for running applications across different systems.